

Integrals – 1

Substitutions: rationalisation, Euler, the half-angle substitution

1 сентября 2026 года

Theoretical minimum.

Def. A substitution reduces an integral to a class we can already handle. Three classes must be recognised instantly: $R(x, x^{p_1/q_1}, \dots)$ is rationalised by $x = t^N$ with N the common denominator of the exponents; $R(x, \sqrt{ax^2 + bx + c})$ is handled by Euler's substitutions; $R(\sin x, \cos x)$ by the half-angle substitution $t = \tan \frac{x}{2}$.

Theorem 1 (Euler's substitutions). Let $y = \sqrt{ax^2 + bx + c}$. If $a > 0$, put $y = \pm\sqrt{a}x + t$; if $c > 0$, put $y = xt \pm \sqrt{c}$; if the trinomial has real roots x_1, x_2 , put $y = t(x - x_1)$. In each case both x and y become rational functions of t .

Theorem 2 (half-angle substitution). For $t = \tan \frac{x}{2}$ one has $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$ and $dx = \frac{2dt}{1+t^2}$. If the integrand is even in the pair $(\sin x, \cos x)$, the substitution $t = \tan x$ is cheaper; if it is odd in $\sin x$, use $t = \cos x$, and symmetrically.

Fact 1. Integration by parts is not a trick but the identity $\int u dv = uv - \int v du$: the only question is what to call dv , and the answer is almost always the most unpleasant factor that can still be integrated.

1. Derive all three Euler substitutions and explain when each of them is the cheapest. Compute $\int \frac{dx}{x\sqrt{x^2+4x+3}}$ and $\int \sqrt{x^2+1} dx$, the second one in two ways (an Euler substitution and a hyperbolic one), and compare the answers.

2. Derive the half-angle formulas and compute $\int \frac{dx}{2+\cos x}$, $\int \frac{dx}{1+\sin x}$ (Integration Bee 2018) and $\int \frac{dx}{\sin x + \tan x}$. Explain why in the third integral $t = \tan \frac{x}{2}$ loses to the substitution $t = \cos x$.
3. (Integration Bee) Rationalising substitutions:
- $\int (1 + x^{1/2} + x^{1/3})(1 + x^{-1/2} + x^{-1/3}) dx$ (2016);
 - $\int \frac{dx}{5 + 4\sqrt{x} + x}$ (2015);
 - $\int \frac{x + \sqrt{x}}{1 + \sqrt{x}} dx$ (2025);
 - $\int \frac{dx}{x + 3\sqrt{x}}$ (2019).
4. (Integration Bee, speed drill) Compute:
- $\int x^2 \sqrt{x^3 + 2} dx$ (2017);
 - $\int x^3 \sqrt{x^2 + 1} dx$ (2016);
 - $\int \frac{x dx}{x^4 + x^2 + 1}$ (2016);
 - $\int \frac{dx}{x(x^2 + 1)}$ (2017);
 - $\int \frac{x^5}{2 + x^{12}} dx$ (2018);
 - $\int \frac{dx}{\sqrt{x}\sqrt{x-x^2}}$ (2018).
5. (Integration Bee) Self-similar integrands: $\int \sqrt{x + \sqrt{x + \sqrt{x + \cdots}}} dx$ (2014), $\int \sqrt{x\sqrt{x\sqrt{x\cdots}}} dx$ (2020) and $\int_0^1 \sqrt{x^2 + x + \sqrt{x^2 + x + \sqrt{\cdots}}} dx$ (2026). First say precisely what the infinite radical means, and only then integrate.